Digital Signal Processing

Zhou Hang

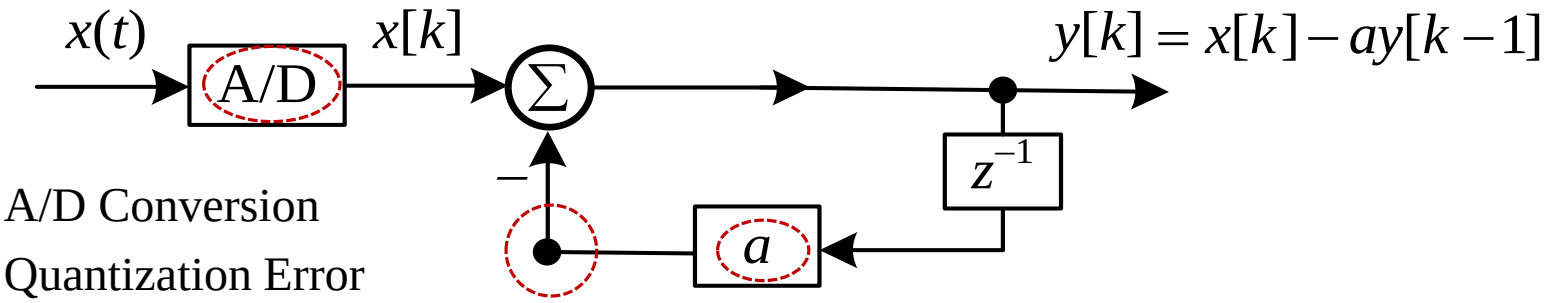
Finite word length effects

- Problem Statement
- Truncation and Rounding Quantization Effects
- Input Signal Quantization Error
- Filter Coefficient Quantization Error
- Product Operation Quantization Error

Problem Statement

In digital systems, the word length of storage units is limited

$$H(z)=\frac{1}{1+az^{-1}}$$



A/D Conversion Quantization Error

Product operation quantization error

Quantization error of filter coefficients

Truncation and rounding quantization effects

- Representation of Fixed-Point Binary Numbers
- Value Quantization and Quantization Error

Representation of fixed-point binary numbers

A fixed-point binary number x has three representations,

$-1<x<1$, its original code, one's complement, and two's complement are defined as follows:

$$[x]_{原}=\begin{cases}0.X_1X_2\cdots X_b & 0\le x<1\\ 1.X_1X_2\cdots X_b & -1\le x<0\end{cases}$$

$$[x]_{反}=\begin{cases}0.X_1X_2\cdots X_b & 0\le x<1\\ 1.\overline{X}_1\overline{X}_2\cdots\overline{X_b} & -1\le x<0\end{cases}$$

$$[x]_{补}=\begin{cases}0.X_1X_2\cdots X_b & 0\le x<1\\ 1.(\overline{X}_1\overline{X}_2\cdots\overline{X_b}+\underbrace{00\cdots0}_{b-1}1) & -1\le x<0\end{cases}$$

The original code, one's complement, and two's complement of a positive number are all the same.

Value quantization and quantization error

Theoretically, any decimal number x $(-1<x<1)$ can be represented with an infinite number of binary digits

$$x = \beta_0 \Delta \sum_{n=1}^{\infty} \beta_n 2^{-n}$$

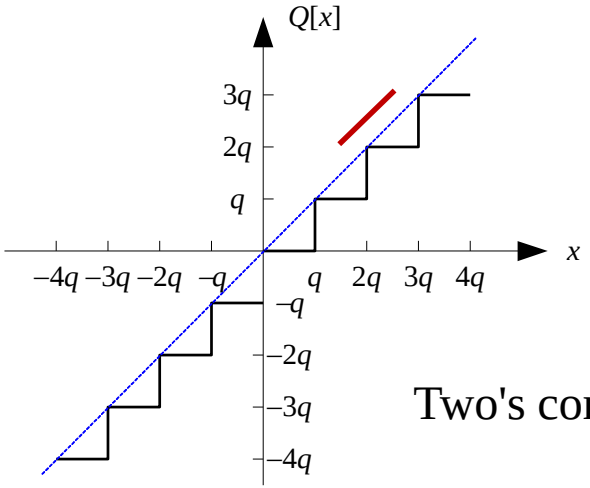
Sign bit — Significant bits

In practice, x can only be approximately represented using (b+1) finite bits, a process known as quantization.

When approximating a number that requires an infinite number of binary digits with a finite-bit binary number, an error is introduced, which is called quantization error.

Value quantization

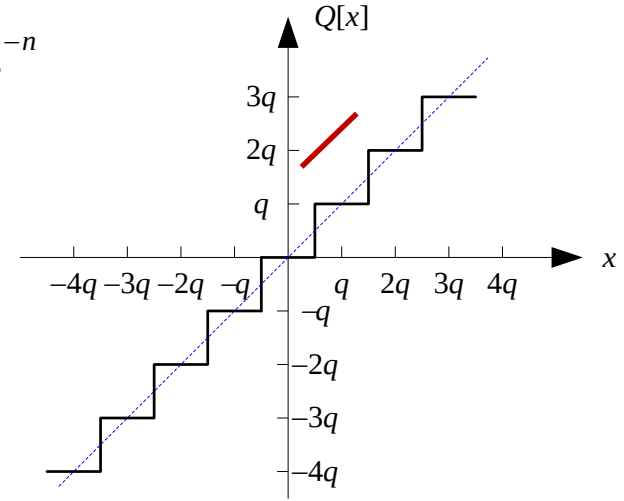
Two types of value quantization methods: Truncation quantization and rounding quantization



$$Q[x] = \beta_0 \, \Delta \sum_{n=1}^{b} \beta_n 2^{-n}$$

$$q = 2^{-b}$$

Two's complement



Truncation quantization

Rounding quantization

Value quantization

The following decimal number sequence:

d=[0.625 0.573 –0.872 0.268 –0.326]

According to the 5-bit truncation sequence: $q = 2^{-5} = 0.03125$

0.62500	0.56250	–0.84375	0.25000	–0.31250
$[0.10100]_b$	$[0.10010]_b$	$[1.11011]_b$	$[0.01000]_b$	$[1.01010]_b$

ccording to the 5-bit rounding sequence: $q = 2^{-5} = 0.03125$

0.62500	0.56250	–0.87500	0.28125	–0.31250
$[0.10100]_b$	$[0.10010]_b$	$[1.11100]_b$	$[0.01001]_b$	$[1.01010]_b$

Value quantization

※ Truncation error: $e = Q[x] - x$

The range of truncation error for positive and negative two's complement numbers is: $-q < e_T \le 0$

$q = 2^{-b}$

The truncation error range for negative numbers in original code and one's complement is: $0 \le e_T < q$

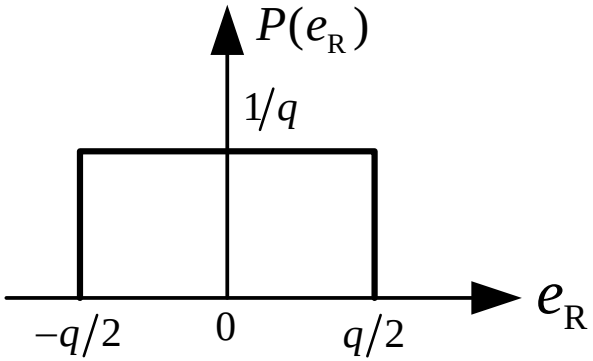
※ Rounding error: $-q/2 < e_R \le q/2$

Rounding error is symmetrically distributed, while truncation error is unipolar.

Value quantization

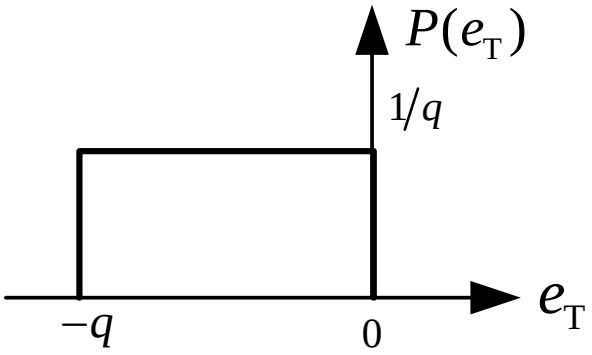
Quantization error (e_R, e_T) statistical analysis: Random variables with uniform distribution.

Rounding error e_R: $-q/2 < e_R \le q/2$



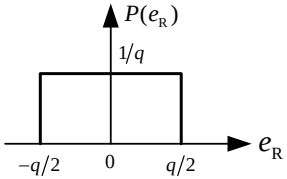
Rounding error probability density function curve

Truncation error e_T: $-q < e_T \le 0$



Truncation error probability density function curve

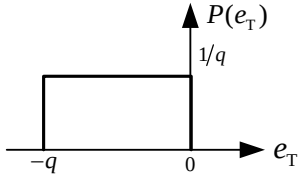
Value quantization



Mean Variance (mean power)

Rounding error e_R: $\mu_R = E\{e_R\} = 0$ $\sigma_R^2 = E\{(e_R - \mu_R)^2\} = \frac{q^2}{12} = \frac{2^{-2b}}{12}$

Truncation error e_T: $\mu_T = E\{e_T\} = -q/2$ $\sigma_T^2 = E\{(e_R - \mu_T)^2\} = \frac{q^2}{12} = \frac{2^{-2b}}{12}$



e_R e_T are the quantization errors at any given moment.

Input signal quantization error

An analog signal is converted into a b-bit digital signal through A/D conversion:

$$\hat{x}[k] = x[k] + e[k]$$

Exact sample value

Quantization error

If the input signal's dynamic range reaches the full scale of the A/D converter and continuously experiences different quantization levels, then the quantization error $e[k]$:

※ $e[k]$ is stationary and follows a uniform distribution, resembling white noise.

※ $e[k]$is uncorrelated with$x[k]$

The statistical characteristics of the quantization error $e[k]$are essentially the same as the quantization error at any given moment(e_R, e_T)

Input signal quantization error

The mean of $e[k]$: $\mu_{e_R} = E\{e[k]\} = 0$

The variance of $e[k]$ (mean power): $\sigma_e^2 = E\{e^2[k]\} = \frac{q^2}{12} = \frac{2^{-2b}}{12}$

The Signal-to-Noise Ratio (SNR) of the input signal is (the average power of the signal $x[k]$ is σ_x^2)

$$\mathrm{SNR} = 10\lg\left(\frac{\sigma_x^2}{\sigma_e^2}\right) \approx 6.02b + 10.79 + 10\lg(\sigma_x^2)\mathrm{dB}$$

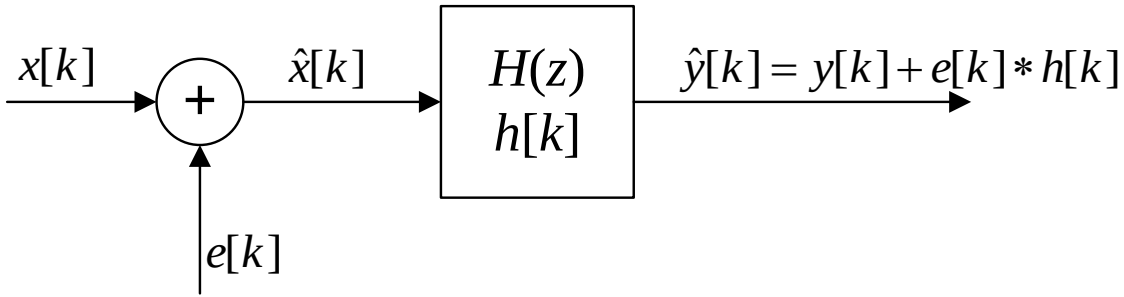
From this, it can be seen that increasing the word length b by one bit increases the SNR by approximately 6 dB.

The quantization error of the input signal passes through an LTI system

The quantized input signal passes through a discrete LTI system, where the input quantization error generates output error at the output end. Based on the mechanism of the input, output, and the system's operation:

$$\hat{y}[k] = \hat{x}[k] * h[k] = \{x[k] + e[k]\} * h[k] = y[k] + e[k] * h[k]$$

Output error



The quantization error of the input signal passes through an LTI system

The output error generated by the input quantization error is

$$v[k] = e[k] * h[k]$$

The variance of the output error $v[k]$ is

$$\sigma_{\mathrm{v}}^{2} = E\{v^{2}[k]\}$$

The variance of the output error $v[k]$ is obtained as

$$\sigma_{\mathrm{v}}^{2} = \sigma_{\mathrm{e}}^{2} \sum_{k=0}^{\infty} h^{2}[k] = \frac{q^{2}}{12} \sum_{k=0}^{\infty} h^{2}[k]$$

The quantization error of the input signal passes through an LTI system

According to Parseval's theorem

$$\sum_{k=0}^{\infty} h^2[k] = \frac{1}{2\pi}\int_{-\pi}^{\pi} \left|H(\mathrm{e}^{\mathrm{j}\Omega})\right|^2 \mathrm{d}\Omega$$

Therefore, the variance of the output error $v[k]$ can also be expressed as

$$\sigma_{\mathrm{v}}{}^2 = \sigma_{\mathrm{e}}{}^2 \frac{1}{2\pi}\int_{-\pi}^{\pi} \left|H(\mathrm{e}^{\mathrm{j}\Omega})\right|^2 \mathrm{d}\Omega$$

Quantization error of the filter coefficients

Due to the limited word length, quantization of the filter coefficients a_n , b_m introduces errors, which result in：

- A deviation between the actual frequency response of the system and the desired system frequency response
- In severe cases, changes in the poles of the system function, which may cause the IIR system to become unstable

※ Quantization Effects in FIR Filter Coefficients

※ Quantization Effects in IIR Filter Coefficients

Quantization Error in FIR Filter Coefficients

$$H(z)=\sum_{m=0}^{M} b_m z^{-m}$$

Quantization of coefficients only affects the zeros and does not involve stability issues but will impact the frequency response. The quantized system function becomes:

$$\hat{H}(z)=\sum_{m=0}^{M} b_m z^{-m}+\sum_{m=0}^{M} \Delta b_m z^{-m}=H(z)+E(z)$$

The frequency response error magnitude $|E(\mathrm{e}^{\mathrm{j}\Omega})|$ is given by:

$$\left|E(\mathrm{e}^{\mathrm{j}\Omega})\right|=\left|\sum_{m=0}^{M}\Delta b_m \mathrm{e}^{-\mathrm{j}m\Omega}\right|\le\sum_{m=0}^{M}\left|\Delta b_m\right|\left|\mathrm{e}^{-\mathrm{j}m\Omega}\right|\le\sum_{m=0}^{M}\left|\Delta b_m\right|$$

Quantization Error in FIR Filter Coefficients

When using rounding quantization, $|\Delta b_m| \le \frac{q}{2}$, therefore

$$\left|E(\mathrm{e}^{\mathrm{j}\Omega})\right| \le \sum_{m=0}^{M} |\Delta b_m| \le \frac{(M+1)q}{2} = (M+1)2^{-(b+1)}$$

If the frequency response error $E(\mathrm{e}^{\mathrm{j}\Omega})$ is specified, then as the order M of the FIR digital filter increases, a larger word length b is required. In practical engineering, it is common to add 3 to 4 extra bits to the estimated word length based on the above formula.

Quantization Error in FIR Filter Coefficients

[Example] Given an FIR digital filter with an order M=28, and the requirement that the frequency response error due to quantization error does not exceed 0.01, determine the required word length.

Solution: According to $\left|E(\mathrm{e}^{\mathrm{j}\Omega})\right| \le (M+1)2^{-(b+1)}$

Substituting the given conditions, we get:

$$(28+1)2^{-(b+1)} \le 0.01$$

$$b \ge 10.50$$

Therefore, the required word length is at least b=11bits.

Quantization Error in FIR Filter Coefficients

[Example] Given a linear phase FIR bandpass filter with the following specifications: $\Omega_{s1}=0.2\pi$ rad, $\Omega_{p1}=0.3\pi$ rad, $\Omega_{p2}=0.6\pi$ rad, $\Omega_{s2}=0.7\pi$ rad, $\delta_p=0.1$, $\delta_s=0.01$ Observe the changes in the system's frequency response when using 4-bit and 10-bit quantized filter coefficients.

```
Fs1=0.2;Fp1=0.3;Fp2=0.6;Fs2=0.7;
f=[Fs1 Fp1 Fp2 Fs2];a=[0 1 0];
Rp=0.1;Rs=0.01;dev=[Rs Rp Rs];
[M,fo,ao,w] = firpmord(f,a,dev);
h = firpm(M,fo,ao,w);
w=linspace(0.01,pi-0.01,1000);
mag=freqz(h,1,w);
plot(w/pi,20*log10(abs(mag)),'r');
```

```
% Coefficient Quantization
num_q1 = qt(h,4);
H = freqz(num_q1, 1, w);
hold on; plot(w/pi,20*log10(abs(H)),'g');
num_q2 = qt(h,10);
H = freqz(num_q2, 1, w);
hold on;
plot(w/pi,20*log10(abs(H)), 'b');
```

Quantization Error in FIR Filter Coefficients

[Example] Given a linear phase FIR bandpass filter with the following specifications: $\Omega_{s1}=0.2\pi$ rad, $\Omega_{p1}=0.3\pi$ rad, $\Omega_{p2}=0.6\pi$ rad, $\Omega_{s2}=0.7\pi$ rad, $\delta_p=0.1$, $\delta_s=0.01$ Quantize the filter coefficients to 4 bits and 10 bits, respectively, and observe the changes in the system's frequency response.

- Unquantized filter coefficients (first 10 data points):

0.0009 0.0070 0.0194 –0.0361 –0.0617 0.0288 0.0573 –0.0013 0.0611 0.0360

- Quantized filter coefficients with 4-bit precision (first 10 data points):

–0.0061 –0.0061 –0.0061 –0.0381 –0.0700 0.0258 0.0258 –0.0061 0.0578 0.0258

- Quantized filter coefficients with 10-bit precision (first 10 data points):

0.0009 0.0069 0.0194 –0.0366 –0.0620 0.0283 0.0568 –0.0016 0.0608 0.0358

Quantization Error in FIR Filter Coefficients

$\Omega_{p1}=0.3\pi$ rad

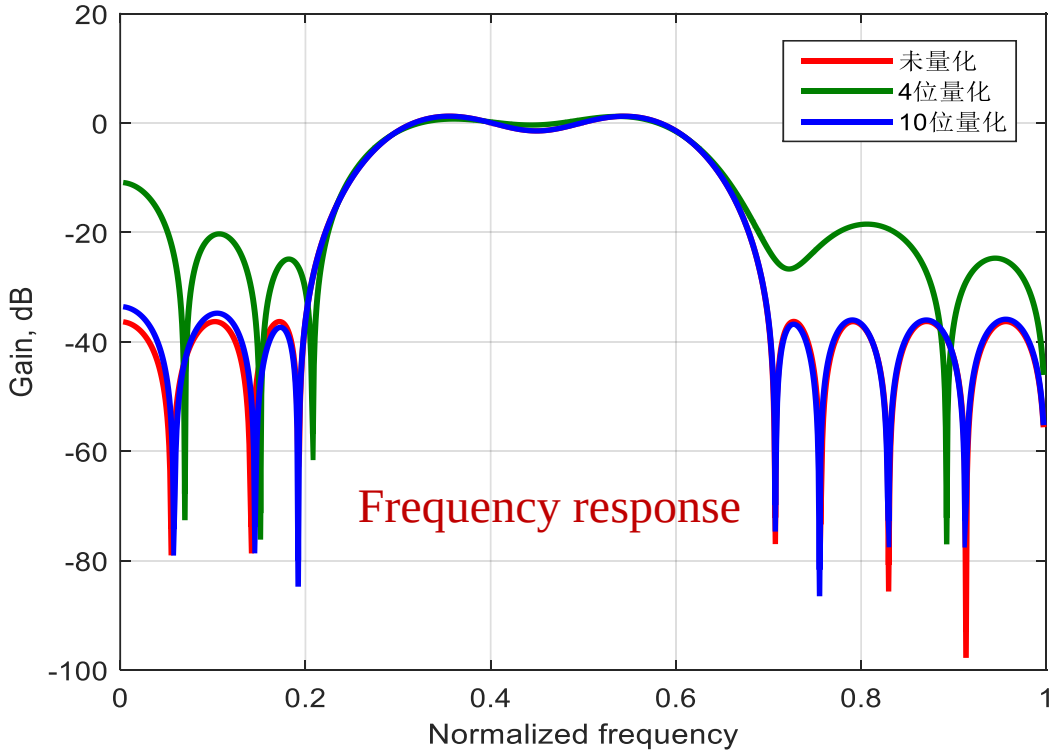
$\Omega_{p2}=0.6\pi$ rad

$\delta_p=0.1$

$\Omega_{s1}=0.2\pi$ rad

$\Omega_{s2}=0.7\pi$ rad

$\delta_s=0.01$



Quantization Error in IIR Filter Coefficients

Impact of Quantization Error in IIR Filter Coefficients on Pole Positions

$$H(z)=\frac{N(z)}{D(z)}=\frac{\sum_{m=0}^{M} b_m z^{-m}}{1+\sum_{n=1}^{N} a_n z^{-n}}=\frac{\sum_{m=0}^{M} b_m z^{-m}}{\prod_{r=1}^{N}(1-p_r z^{-1})}$$

Quantized values of coefficients a_n: $\hat{a}_n = a_n + \Delta a_n, \quad n = 1, \cdots, N$

Quantized values of poles p_r: $\hat{p}_r = p_r + \Delta p_r, \quad r = 1, \cdots, N$

Quantization Error in IIR Filter Coefficients

The impact of quantization error in IIR filter coefficients on pole positions

Quantized values of poles p_r:

$$\hat{p}_r = p_r + \Delta p_r, \quad r = 1, \cdots, N$$

Quantization error of poles

$$\sum_{n=1}^{N} \frac{\partial p_r}{\partial a_n} \Delta a_n$$

The sensitivity of the *r-th* pole to the quantization of the *n-th* coefficient a_n

The larger $\partial p_r / \partial a_n$, the greater the impact of Δa_n on Δp_r, and vice versa.

Quantization Error in IIR Filter Coefficients

The expression for the sensitivity of pole positions is given by:

$$D(z) = 1 + \sum_{n=1}^{N} a_n z^{-n} = \prod_{r=1}^{N} (1 - p_r z^{-1})$$

$$\left.\frac{\partial p_r}{\partial a_n}\right|_{z=p_r} = \frac{\left.\frac{\partial D(z)}{\partial a_n}\right|_{z=p_r}}{\left.\frac{\partial D(z)}{\partial p_r}\right|_{z=p_r}} \frac{p_r^{-n}}{-p_r^{-1}\prod_{\substack{l=1 \\ l\neq r}}^{N}(1 - p_l p_r^{-1})} = \frac{-p_r^{N-n}}{\prod_{\substack{l=1 \\ l\neq r}}^{N}(p_r - p_l)}$$

- The greater the distance between poles, the lower the sensitivity of pole positions; conversely, the closer the poles are to each other, the higher the sensitivity of pole positions.
- In parallel and cascade structures, each subsystem has at most two conjugate poles, so the quantization impact is relatively small.

Quantization Error in IIR Filter Coefficients

[Example] Design a band-stop filter with the following specifications and implement it using a direct-form structure. Quantize the filter coefficients to 4-bit, 8-bit, and 12-bit precision, and observe the changes in the filter's frequency response and zero-pole locations.

$$\Omega_{p1}=0.45\pi \text{ rad}, \quad \Omega_{p2}=0.72\pi \text{ rad}, \quad A_p \le 1\text{dB},$$
$$\Omega_{s1}=0.52\pi \text{ rad}, \quad \Omega_{s2}=0.62\pi \text{ rad}, \quad A_s \ge 20\text{dB}$$

Quantization Error in IIR Filter Coefficients

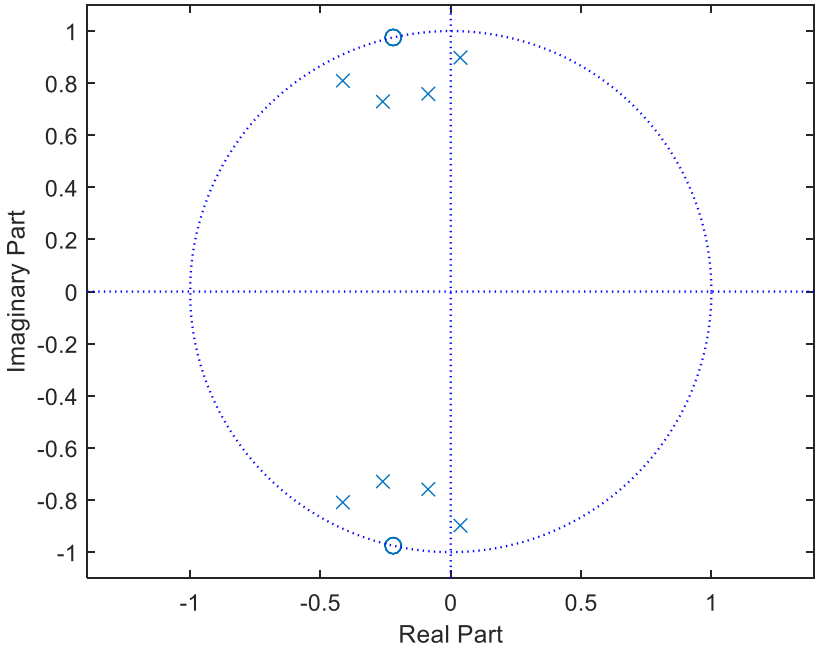
Unquantized filter coefficients:

0.4832	0.8537	2.4983	2.7277	4.0487	2.7277	2.4983	0.8537	0.4832	Numerator coefficients b_i
1.4521	3.3737	3.0764	3.7549	2.1484	1.6497	0.4860	0.2335		Denominator coefficients a_i

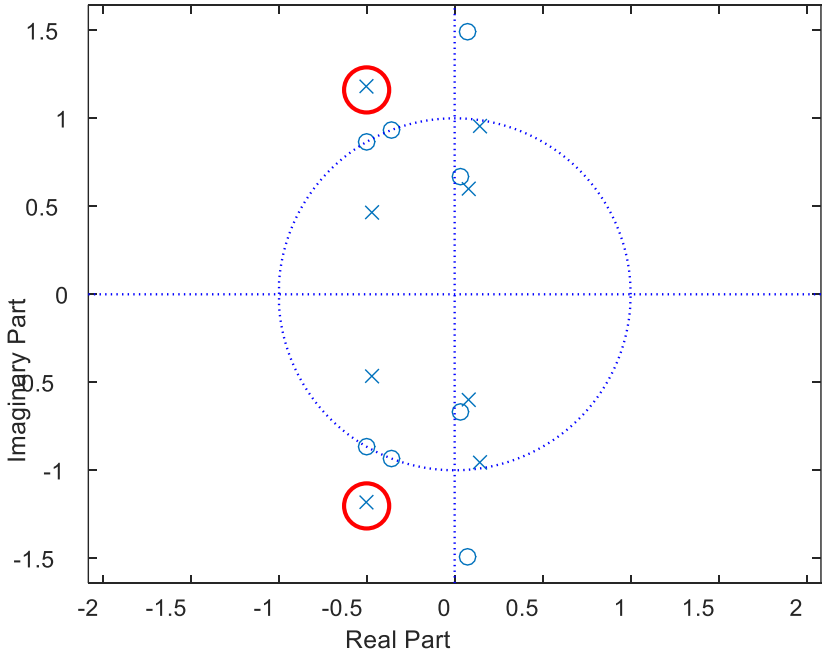
Filter coefficients after 4-bit quantization:

0.4719	0.7104	2.3796	2.6180	4.0487	2.6180	2.3796	0.7104	0.4719	Numerator coefficients b_i
1.4257	3.3334	2.8565	3.5718	2.1411	1.4257	0.4719	0.2335		Denominator coefficients a_i

Quantization Error in IIR Filter Coefficients



Unquantized



4-bit quantization Unstable system

Quantization Error in IIR Filter Coefficients

Unquantized filter coefficients:

0.4832	0.8537	2.4983	2.7277	4.0487	2.7277	2.4983	0.8537	0.4832
1.4521	3.3737	3.0764	3.7549	2.1484	1.6497	0.4860	0.2335	

Numerator coefficients b_i

Denominator coefficients a_i

Filter coefficients after 8-bit quantization:

0.4719	0.8445	2.4839	2.7223	4.0487	2.7223	2.4839	0.8445	0.4719
1.4406	3.3632	3.0651	3.7507	2.1411	1.6493	0.4719	0.2335	

Numerator coefficients b_i

Denominator coefficients a_i

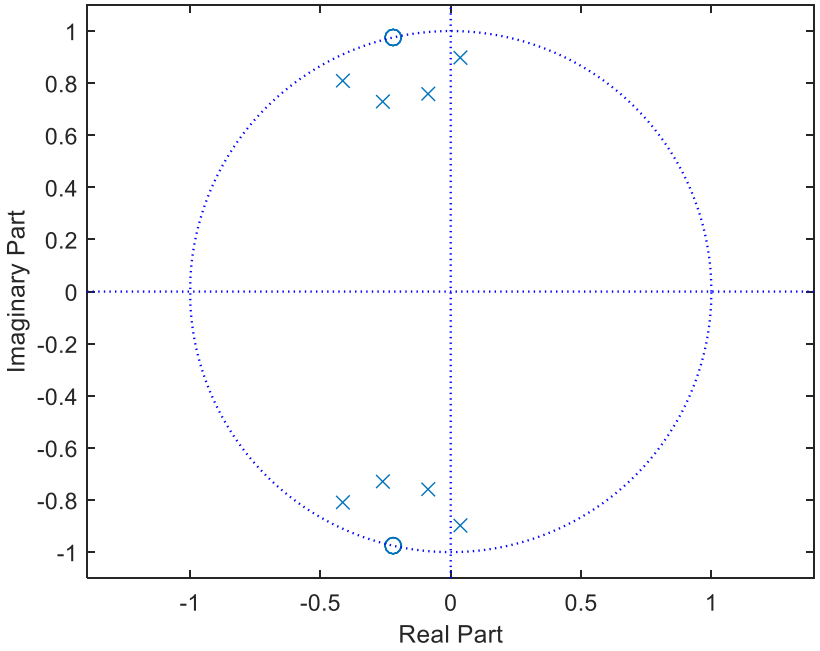
Filter coefficients after 12-bit quantization:

0.4831	0.8529	2.4979	2.7270	4.0487	2.7270	2.4979	0.8529	0.4831
1.4518	3.3734	3.0763	3.7544	2.1476	1.6493	0.4859	0.2335	

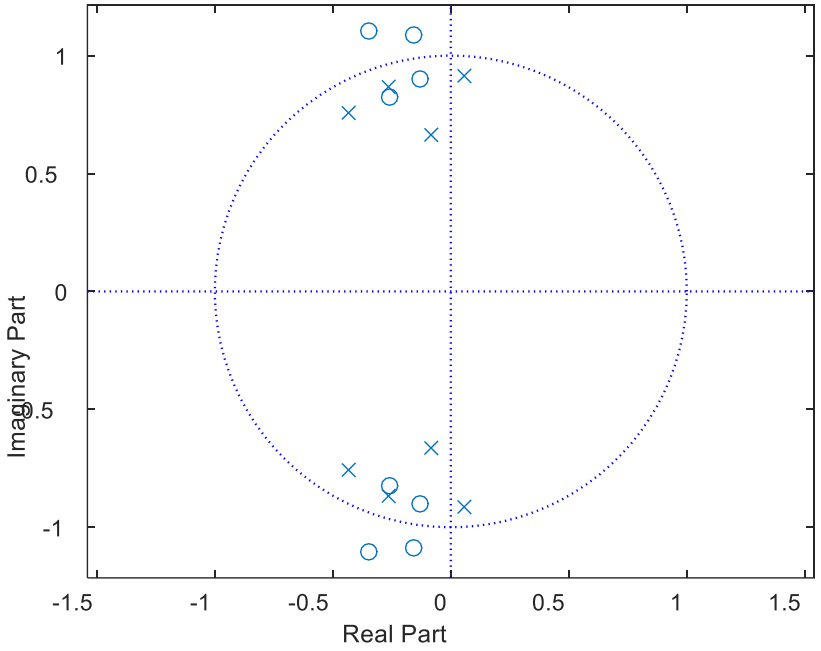
Numerator coefficients b_i

Denominator coefficients a_i

Quantization Error in IIR Filter Coefficients

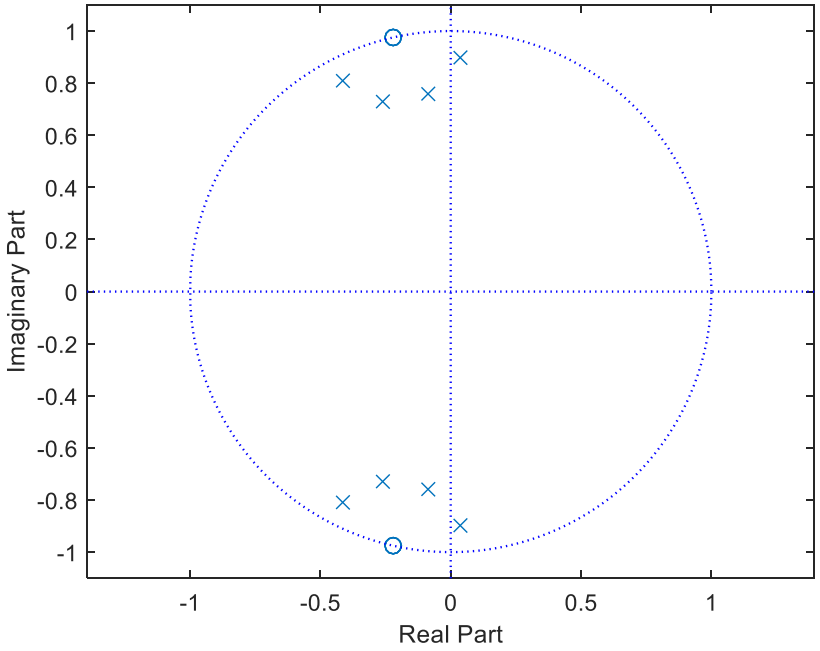


Unquantized

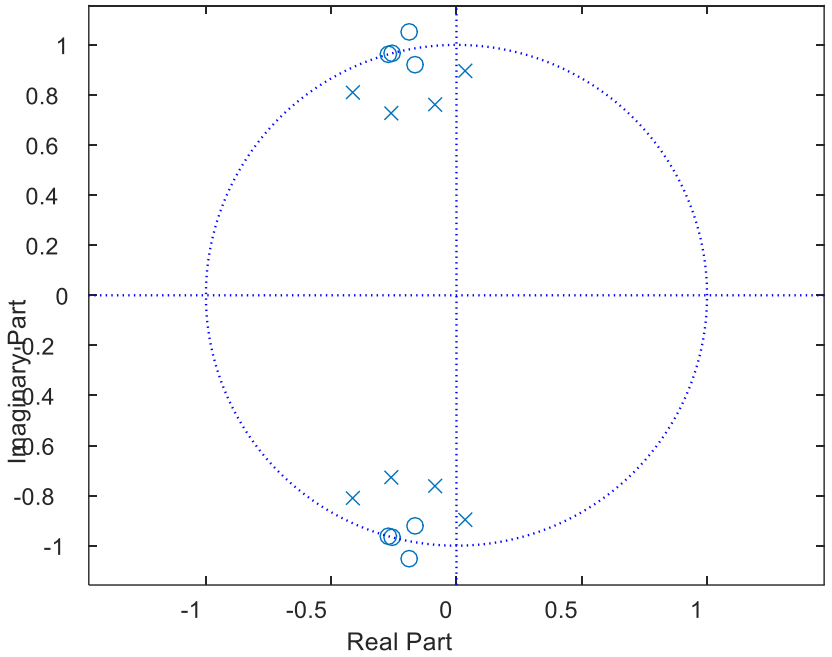


8-bit quantization

Quantization Error in IIR Filter Coefficients



Unquantized

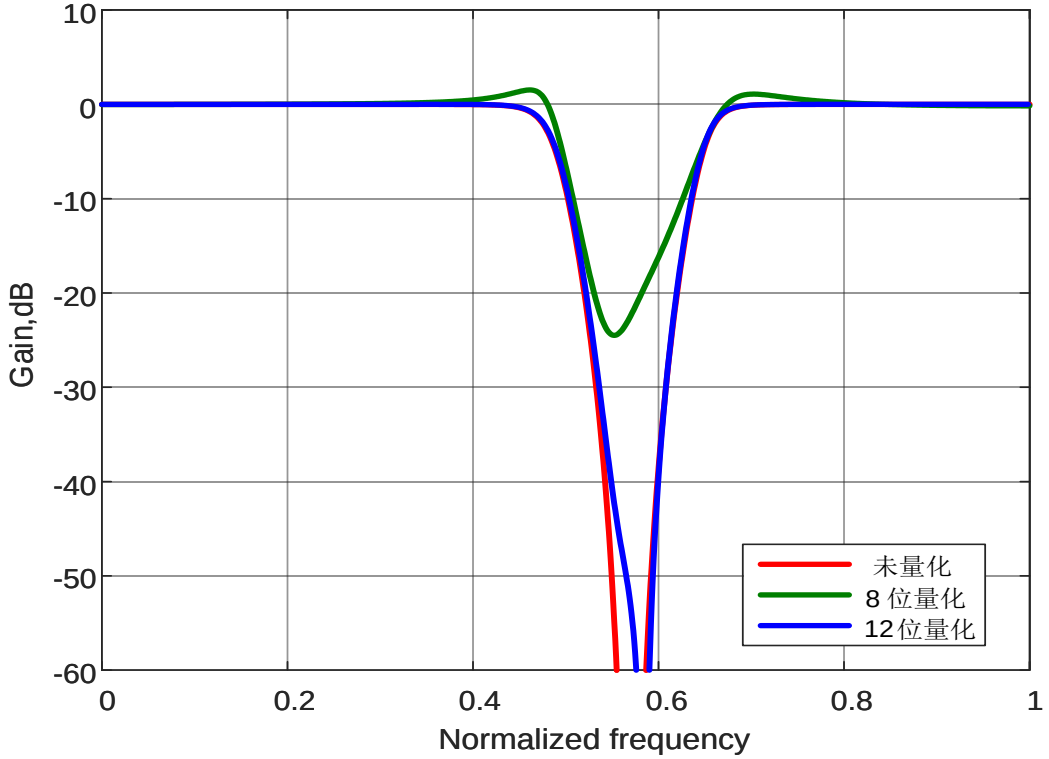


12-bit quantization

Quantization Error in IIR Filter Coefficients

Ω_{p1} =0.45π rad
Ω_{p2} =0.72π rad
A_p≤1dB

Ω_{s1} =0.52π rad
Ω_{s2} =0.62π rad
A_s ≥20dB

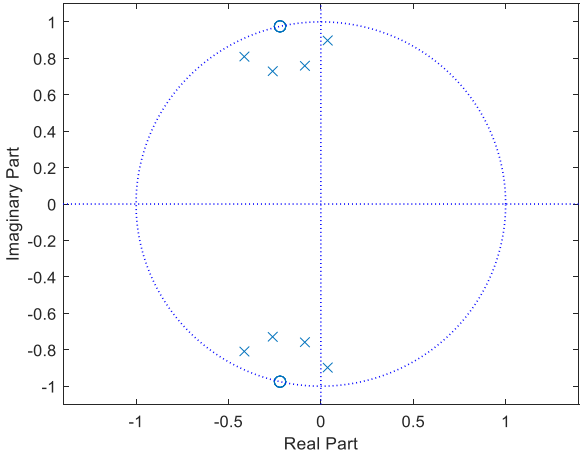


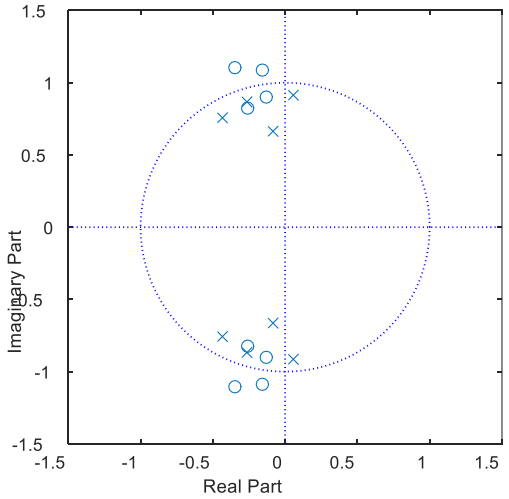
Quantization Error in IIR Filter Coefficients

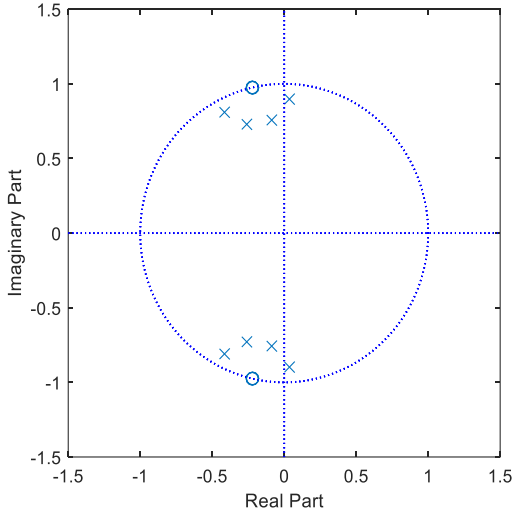
[Example] Design a band-stop filter with the following specifications. If the filter is implemented using a cascade of second-order sections and the coefficients are still quantized to 8-bit precision, compare the impact on the filter's frequency response and zero-pole locations with that of a direct-form structure.

$$\Omega_{p1}=0.45\pi \text{ rad}, \quad \Omega_{p2}=0.72\pi \text{ rad}, \quad A_p \le 1\text{dB},$$
$$\Omega_{s1}=0.52\pi \text{ rad}, \quad \Omega_{s2}=0.62\pi \text{ rad}, \quad A_s \ge 20\text{dB}$$

Quantization Error in IIR Filter Coefficients







Direct-form (unquantized) Direct-form (8-bit quantization) Cascade form (8-bit quantization)

Quantization Error in IIR Filter Coefficients

Ω_{p1} =0.45π rad

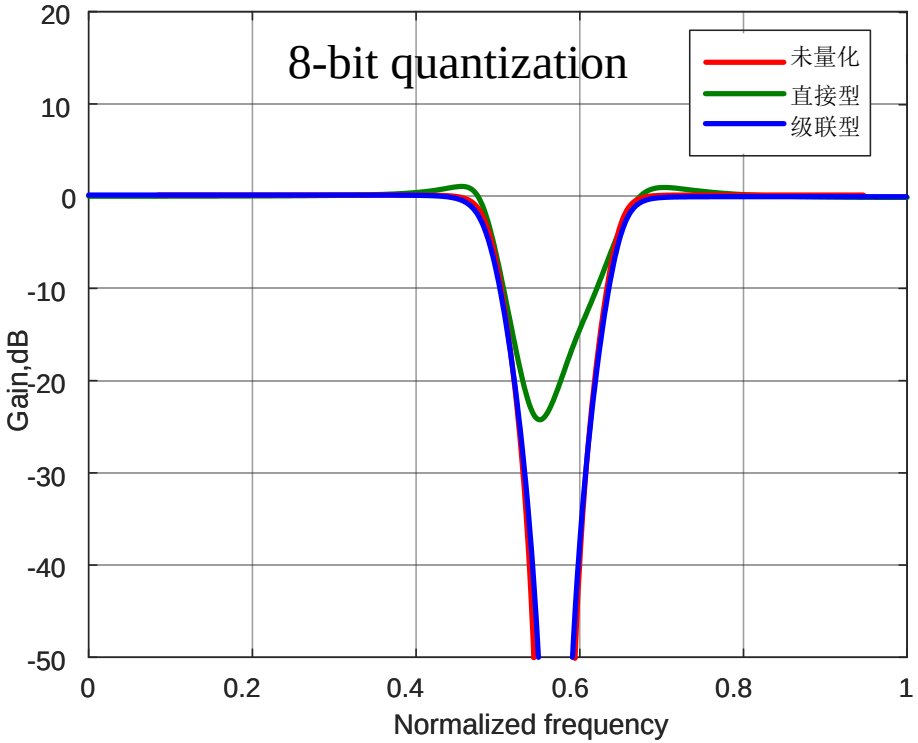
Ω_{p2} =0.72π rad

$A_p \leq 1$dB

Ω_{s1} =0.52π rad

Ω_{s2} =0.62π rad

$A_s \geq 20$dB



Implementing a digital filter using MATLAB

```
Wp=[0.45,0.72];Ws=[0.52,0.62]; Ap=1;As=20; B = 8; % Number of quantization bits
w=linspace(0,pi,1024);
[N,Wc]=buttord(Wp,Ws,Ap,As);
[numd,dend]=butter(N,Wc,'stop');
% Quantize the coefficients a and b of the direct-form structure
len1 = length(numd); len2 = length(dend);
pa = [numd dend(2:end)];                % Numerator b and denominator a coefficients
pa_q = qt(pa,B);                        % Coefficient quantization
num = pa_q(1:len1);                     % Quantized numerator
den = [1 pa_q(len1+1:len1+len2-1)];     %Quantized denominator, with a 1 added in front of the coefficient a
H_direct = freqz(num, den, w);
figure; plot(w/pi,20*log10(abs(H_direct )),'r');
```

Implementing a digital filter using MATLAB

```
sos = tf2sos(numd, dend); % Convert direct-form to cascade form
num2 = sos(1, 1:3); den2 = sos(1, 4:6);   % Numerator and denominator of the first subsystem in the cascade form
pa = [num2 den2(2:end)]; pa_q = qt(pa,B);   %Quantize the numerator and denominator coefficients, where the denominator coefficients are a
num2 = pa_q(1:3);den2 = [1 pa_q(4:5)]; % Quantized numerator and denominator
H2 = freqz(num2, den2, w); % Frequency response of the first subsystem in the cascade form
H = abs(H2);
for i = 2:N
num2 = sos(i, 1:3); den2 = sos(i, 4:6);     %Numerator and denominator of the i-th subsystem in the cascade form
    pa = [num2 den2(2:end)]; pa_q = qt(pa,B);
    num2 = pa_q(1:3);den2 = [1 pa_q(4:5)];
    H2 = freqz(num2, den2, w);            % Frequency response of the i-th subsystem in the cascade form
H = H.*abs(H2);                        % Multiply the frequency responses of the cascade form subsystems
end
plot(w/pi,20*log10(abs(H)),'b');
```

Implementing a digital filter using MATLAB

```
% qt: Quantization of sequences
% input   a: Sequences that need to be quantized，B: Quantization bit depth
% output aq: Quantized sequence
function aq = qt(a,B)
max_v = max(a);
min_v = min(a);
a = (a-min_v)/(max_v-min_v);
a = fix(a*(2^B));
a = a/(2^B);
aq = a*(max_v-min_v)+min_v;
end
```

Quantization error in multiplication operations

- Statistical Analysis of Quantization Error in FIR System Multiplication
- Statistical Analysis of Quantization Error in IIR System Multiplication
- Limit cycle oscillation phenomenon in IIR systems

Quantization error in multiplication operations

The product quantization error is represented by noise sources $e_i[k]$ with the following assumptions:

（1）Each noise source is a white noise sequence.

（2）The noise sources are statistically independent and uncorrelated with each other.

（3）Within the quantization noise range, each noise source is considered to have an equal probability density distribution.

The variance of the product quantization noise source is:

$$\sigma^2 = \frac{q^2}{12} = \frac{2^{-2b}}{12}$$

Statistical Analysis of Quantization Error in FIR System Multiplication

The difference equation for an *M-th* order FIR system is:

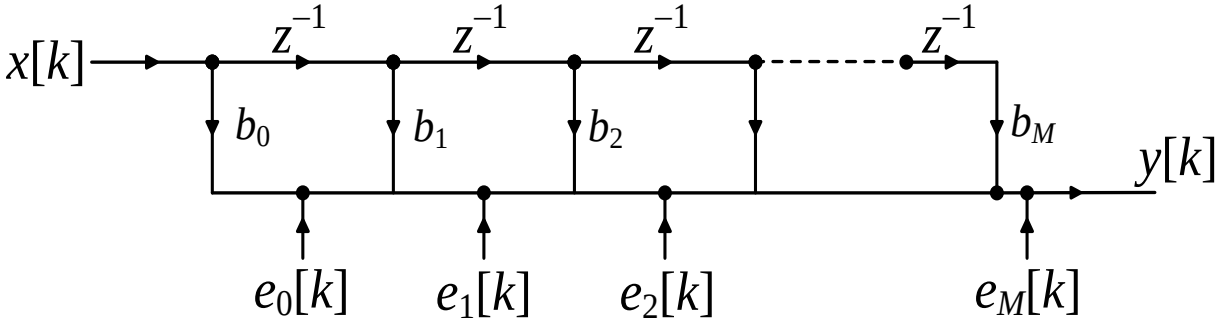
$$y[k] = \sum_{m=0}^{M} b_m x[k-m]$$

Quantize the M+1 multiplication terms in an FIR system.

$$\hat{y}[k] = \sum_{m=0}^{M} \{b_m x[k-m] + e_m[k]\} = y[k] + \sum_{m=0}^{M} e_m[k]$$

Statistical Analysis of Quantization Error in FIR System Multiplication

Noise source model for product quantization error in a direct-form FIR structure:



In a direct-form FIR structure, all the noise is added directly at the output.

The output noise variance generated by product quantization noise

$$\sigma_o^2 = (M+1)\sigma^2 = \frac{(M+1)}{12} q^2$$

Statistical Analysis of Quantization Error in IIR System Multiplication

Difference equation of the IIR system

$$y[k] = \sum_{m=0}^{M} b_m x[k-m] - \sum_{n=1}^{N} a_n y[k-n]$$

Quantize the $M+1+N$ multiplication terms in the above equation

$$\hat{y}[k] = \sum_{m=0}^{M} \{b_m x[k-m] + e_m[k]\} - \sum_{n=1}^{N} \{a_n y[k-n] + e_n[k]\}$$

Statistical Analysis of Quantization Error in IIR System Multiplication

[Example] Given the system function of an IIR digital filter as

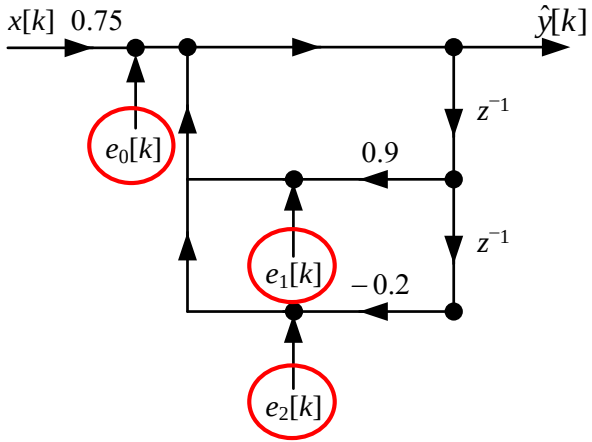
$$H(z) = \frac{0.75}{(1-0.5z^{-1})(1-0.4z^{-1})}$$

calculate the output noise variance due to multiplication quantization for direct-form, cascade form, and parallel form structures.

Statistical Analysis of Quantization Error in IIR System Multiplication

Solution: Direct-form Structure

$$H(z)=\frac{0.75}{(1-0.5z^{-1})(1-0.4z^{-1})}=\frac{0.75}{1-0.9z^{-1}+0.2z^{-2}}$$



The system through which the quantization error $e[k]$ passes is

$$H_e(z)=\frac{1}{1-0.9z^{-1}+0.2z^{-2}}=\frac{5}{1-0.5z^{-1}}-\frac{4}{1-0.4z^{-1}}$$

$$h_e[k]=\mathcal{Z}^{-1}\{H_e(z)\}=[5(0.5)^k-4(0.4)^k]u[k]$$

Statistical Analysis of Quantization Error in IIR System Multiplication

The variance of the quantization error $e[k]$(including $e_0[k]$、$e_1[k]$ and $e_2[k]$)is

$$\sigma_e^2 = 3 \times \frac{q^2}{12} = \frac{q^2}{4}$$

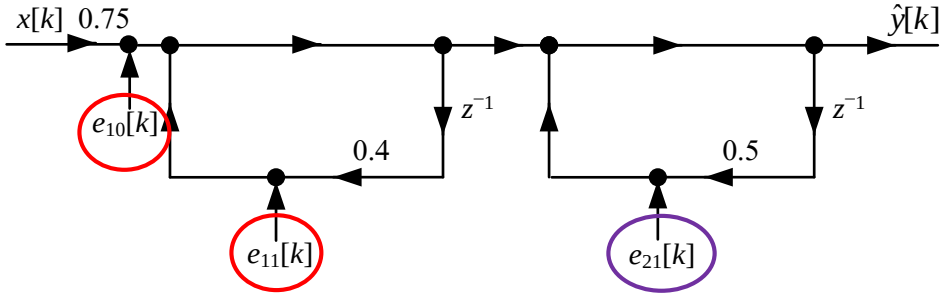
Variance of the output noise produced by multiplication quantization

$$\sigma_o^2 = \sigma_e^2 \sum_{k=0}^{\infty} h_e^2[k] = \frac{q^2}{4} \sum_{k=0}^{\infty} [5(0.5)^k - 4(0.4)^k]^2 = 0.5952q^2$$

Statistical Analysis of Quantization Error in IIR System Multiplication

Cascade Form Structure

$$H(z) = 0.75 \times \left(\frac{1}{1-0.4z^{-1}}\right)\left(\frac{1}{1-0.5z^{-1}}\right)$$



System through which $e_1[k](e_{10}[k]$ and $e_{11}[k])$ passes:

$$H_{e_1}(z) = \left(\frac{1}{1-0.4z^{-1}}\right)\left(\frac{1}{1-0.5z^{-1}}\right)$$

System through which $e_2[k\,](e_{21}[k])$ passes:

$$H_{e_2}(z) = \frac{1}{1-0.5z^{-1}}$$

Statistical Analysis of Quantization Error in IIR System Multiplication

The system through which $e_1[k]$ ($e_{10}[k]$ and $e_{11}[k]$) passes:

$$h_{e_1}[k] = [5(0.5)^k - 4(0.4)^k]u[k]$$

The system through which $e_2[k]$ ($e_{21}[k]$) passes:

$$h_{e_2}[k] = (0.5)^k u[k]$$

Variance of the output noise produced by multiplication quantization

$$\sigma_o^2 = \sigma_{e_1}^2 \sum_{k=0}^{\infty} h_{e_1}^2[k] + \sigma_{e_2}^2 \sum_{k=0}^{\infty} h_{e_2}^2[k]$$

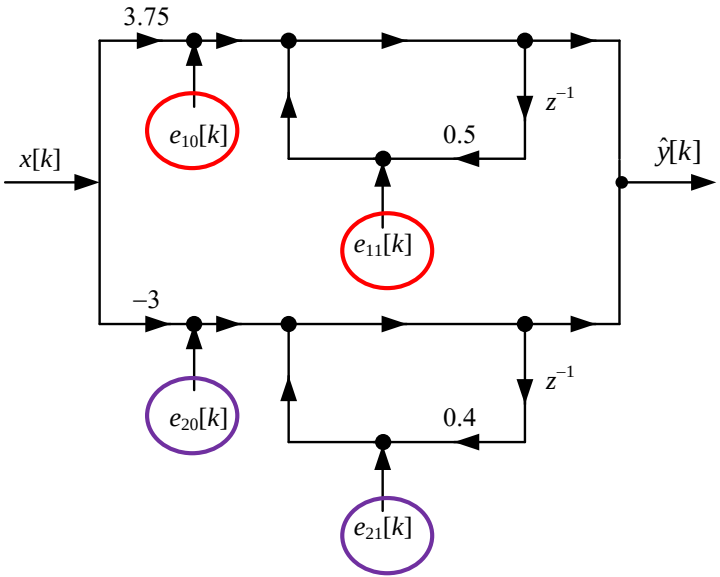
$$\sigma_{e_1}^2 = 2\frac{q^2}{12},\ \sigma_{e_2}^2 = \frac{q^2}{12}$$

$$= 2\frac{q^2}{12}\sum_{k=0}^{\infty} h_{e_1}^2[k] + \frac{q^2}{12}\sum_{k=0}^{\infty} h_{e_2}^2[k] = 0.3968q^2 + 0.1111q^2 = 0.5079q^2$$

Statistical Analysis of Quantization Error in IIR System Multiplication

Parallel Form Structure

$$H(z) = \frac{3.75}{1-0.5z^{-1}} + \frac{-3}{1-0.4z^{-1}}$$



System through which $e_1[k](e_{10}[k]$ and $e_{11}[k])$ passes:

$$H_{e_1}(z) = \frac{1}{1-0.5z^{-1}}$$

System through which $e_2[k\ ](e_{21}[k])$ passes:

$$H_{e_2}(z) = \frac{1}{1-0.4z^{-1}}$$

Statistical Analysis of Quantization Error in IIR System Multiplication

The system through which $e_1[k]$ ($e_{10}[k]$ and $e_{11}[k]$) passes:

$$h_{e_1}[k] = (0.5)^k u[k]$$

The system through which $e_2[k]$ ($e_{21}[k]$) passes:

$$h_{e_2}[k] = (0.4)^k u[k]$$

Variance of the output noise produced by multiplication quantization

$$\sigma_o^2 = \sigma_{e_1}^2 \sum_{k=0}^{\infty} h_{e_1}^2[k] + \sigma_{e_2}^2 \sum_{k=0}^{\infty} h_{e_2}^2[k]$$

$$= 2\frac{q^2}{12}\{\sum_{k=0}^{\infty} h_{e_1}^2[k] + \sigma_{e_2}^2 \sum_{k=0}^{\infty} h_{e_2}^2[k]\} = 0.4206q^2$$

$$\sigma_{e_1}^2 = 2\frac{q^2}{12}, \sigma_{e_2}^2 = 2\frac{q^2}{12}$$

Statistical Analysis of Quantization Error in IIR System Multiplication

Variance of the output noise produced by multiplication quantization

Direct-form Structure: $\sigma_o^2 = 0.5952q^2$

Cascade Form Structure: $\sigma_o^2 = 0.5079q^2$

Parallel Form Structure: $\sigma_o^2 = 0.4206q^2$

Statistical Analysis of Quantization Error in IIR System Multiplication

Comparative Analysis of Product Quantization Errors in Different Structures:

- Direct-form Structure: All product quantization errors must pass through the entire feedback path.
- Cascade Structure: Each error passes through only the subsequent feedback paths and not the preceding ones, but the errors are dependent on the order of the cascade.
- Parallel Structure: Each error in a parallel network is only related to the feedback path of its own branch and is independent of the other parallel networks.

Limit Cycle Oscillation Phenomenon in IIR Systems

For a stable IIR digital filter, if the computational precision is infinite, when $n>n_0$and the input stops, the filter's output will gradually decay to zero as $n>n_0$However, if computations are performed with finite word length, the filter's output, after $n>n_0$, may decay to a certain non-zero amplitude range and then exhibit oscillatory behavior. This phenomenon is known as zero-input limit cycle oscillation.

Limit Cycle Oscillation Phenomenon in IIR Systems

For a first-order difference equation:

$$y[k]-\alpha y[k-1]=x[k]$$

Assume $y[-1]=0$, word length $b=3$, $\alpha=1/2=[0.100]_b$, $x[k]=(7/8)\delta[k]=[0.111]_b\,\delta[k]$

For infinite precision computation, the output of the IIR system is:

$$y[k]=\frac{7}{8}\left(\frac{1}{2}\right)^k u[k]$$

The difference equation corresponding to the product quantization is:

$$\hat{y}[k]=x[k]+Q\{\alpha\,\hat{y}[k-1]\}$$

Limit Cycle Oscillation Phenomenon in IIR Systems

When $y[-1]=0$, word length $b=3$, $\alpha=1/2=[0.100]_b$, $x[k]=(7/8)\delta[k]=[0.111]_b\,\delta[k]$

$$\hat{y}[0]=x[0]+Q\{\alpha\,\hat{y}[-1]\}=[0.111]_b=7/8$$

$$\hat{y}[1]=Q\{\alpha\,\hat{y}[0]\}=Q[0.0111]=[0.100]_b=4/8$$

$$\hat{y}[2]=Q\{\alpha\,\hat{y}[1]\}=Q[0.010]=[0.010]_b=2/8$$

$$\hat{y}[3]=Q\{\alpha\,\hat{y}[2]\}=Q[0.001]=[0.001]_b=1/8$$

$$\hat{y}[4]=Q\{\alpha\,\hat{y}[3]\}=Q[0.0001]=[0.001]_b=1/8$$

……

For finite word length computation, the output of the IIR system is:

$$\hat{y}[k]=\{\frac{7}{8},\frac{4}{8},\frac{2}{8},\frac{1}{8},\frac{1}{8},\frac{1}{8},\ldots\}$$

limit cycle oscillation

Thanks!

The materials referenced in this course are accumulated over years of teaching by the instructor, sourced from various media and exchanges with colleagues and peers. It is difficult to indicate the source of each material one by one. This is to clarify and express gratitude!